DPUSGEInstrumentPanel

Inserts an instrument panel (e.g. a dashboard) into the visualization SGE.

Up to ten lights, symbols or bar-type displays can be shown on the instrument panel. Round gauges can be visualized using additional instances of DPUSGE-RoundGauge.

The instrument panel is shown two-dimensionally, on top of the 3D environment.

Tip: If a separate monitor is used for the instrument panel, and it is not required to insert it into the main SGE view, it is recommended to use DPUMMI instead, which provides additional types of display elements.

1. Dynamic parameters for the background:

X, Y	Position of the instrument panel in SILAB's overlay coor-dinate system (see "Reference Visualization SGE" for a description of coordinate systems)
W, H	Width and height of the instrument panel in pixels
Z	Z position for the instrument panel (0..10, where 10 is in the front). Default: 5
Show	Current visibility of the instrument panel. Default: *true*

2. Static parameters for the background:

TexInstrument	SGE resource name of the background texture (e.g. *"textures.hud.bg_panel"*)
Visibility	Visibility category for the object. It can be used to display the object on one of the views updated on a single graphics computer only.

3. Inputs for each instrument:

In.Value	Current value of the displayed quantity.

4. Dynamic parameters for each instrument:

In.X, In.Y	Position of the instrument within the panel. Range of values: 0..1. 0 is to the left and top, 1 is to the bottom and right.
In.W, In.H	Width and height of the instrument, relative to the panel Range of values: 0..1. 1 is the total size of the instrument panel.
In.ValueMin	Minimum possible input value (*Value*) Default: 0
In.ValueMax	Maximum possible input value (*Value*) Default: 1
In.Show	Is this instrument currently visible?

	The instrument n is only visible if Show and *In.Show* are 1 at the same time.

5. Static parameters for each instrument (n = 1..10):

In.TexInstrument	SGE resource name for the background of this instrument (e.g. "*tex-tures.hud.bg_lamp"*). If *TexInstrument* has been specified, this back-ground is painted independently of the *Value* input.
In.TexOverlay	Resource name for the display element (e.g. glowing lamp). *TexO-verlay* is only painted on top of *TexInstrument*, when Value is greater than 0.
In.OverlayParts	Number of alternative symbols that are contained in the image file *TexOverlay*. The default value is 1, i.e. the image contains only one symbol. If *OverlayParts* > 1, the symbols must be of same size and aligned from left to right within the image (see example below).
In.OverlayDir	Only relevant for bar-type displays (e.g. fuel tank fill level). In this case, *OverlayParts* must be 1. Depending on Val-ue, only a part of *TexOverlay* is painted. If Value = *ValueMin*, *TexOverlay* is not visible at all; for Value = *Val-ueMax*, it is completely visible. For values in-between, the image is partially visible: **OverlayDir** — **Richtung** 1 — lower part 2 — left part 3 — upper part 4 — right part
In.BlinkPeriodMS	If set to > 0, the symbol *TexOverlay* blinks in this interval (while *Value* is constantly 1). The default setting is 0, which means that the overlay isn't blinking.

6. Dependencies to other DPUs:

DPUSGEInstrumentPanel depends on **DPUSGEWorld** and **DPUSGEView**, which must also run on the computer. The DPU's Index values must conform to the follow-ing equation:

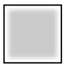
SGEWorld.Index < SGERoundGauge.Index < SGEView.Index

7. Functional principle:

The following script simulates a dashboard with embedded left indicator, headlamp and tank fill level.

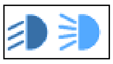
```
DPUSGEInstrumentPanel Inst
{
	Computer = {VIS};
	Index = 50;

	TexBackground = "hud.bg_panel_middle"; # grey background

	X = 0;
	Y = 256;
```

```
    W = 256;
    H = 256;
    Z = 8;

    # Left indicator ################

    I1.TexInstrument = "hud.bg_lamp_transparent"; # image:
```



```
    I1.X = 0.21;
    I1.Y = 0.11;
    I1.W = 0.2;
    I1.H = 0.2;

    I1.TexOverlay = "hud.indicator_left"; # image:
```

```
    I1.BlinkPeriodMS = 1000;      # Automatic blinking

    # Headlamp ####################
    I2.TexInstrument = "hud.bg_lamp_transparent";
    I2.X = 0.475;
    I2.Y = 0.11;
    I2.W = 0.2;
    I2.H = 0.2;

    I2.TexOverlay = "hud.light"; # image:
```



```
    I2.OverlayParts = 2;    # “lower beam” and “upper beam”
    I2.ValueMax = 2;        # 0 = lights off, 1 = lower beam, 2 = upper

    # Tank fill level ####################
    I3.TexInstrument = "hud.bg_lamp_transparent";
    I3.X = 0.21;
    I3.Y = 0.70;
    I3.W = 0.40;
    I3.H = 0.10;
    I3.TexOverlay = "color.red";
    I3.OverlayDir = 2;
    I3.ValueMax = 80;       # Tank capacity 80 liters
};
```

The input values I1.Value = 1, I2.Value = 1, I3.Value = 55 result in the following display: